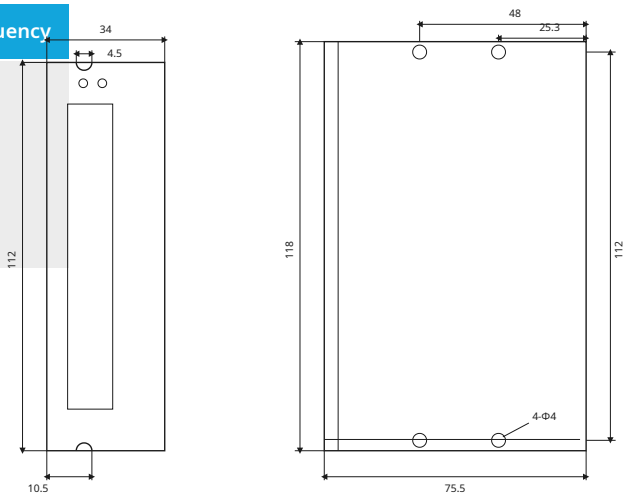
DM Series Two-Phase Stepper Driver FYDM504T

English translation of the XFOYO (兴丰元) FYDM504T product page datasheet · source: xfoyo.com/a/bujinqudong/liangxiangbujinqudongqi/417.html

Product Overview

Model	Voltage	Drive current	Subdivision	Max. pulse frequency
FYDM504T	DC24~50V AC20~40V	1.0 ~ 4.5 A/phase 8 settings, adjustable	Max. resolution 25600 steps/rev 16 settings, adjustable	250 Kpps

Product Mounting Dimensions



Units: mm. Left: side view — 34 wide × 112 high, housing notch 4.5 wide, 10.5 from the edge. Right: front view — 75.5 wide × 118 high, four Φ4 mounting holes; the hole rows are 112 mm apart, and the two hole columns are 48 mm and 25.3 mm from the right-hand edge (22.7 mm apart).



Product Features

- **High performance, high efficiency:** the latest 32-bit DSP control technology, so the motor reaches its best output performance.
- **Low vibration, low noise:** automatically calculates the resonance point and effectively suppresses the motor's noise and vibration at low running speeds.
- **Small temperature rise, long life:** precision current control keeps the motor's temperature rise low.
- **Wide speed range:** with a suitable stepper motor, up to 1500 rpm.
- **Interference frequency filtering:** optically isolated differential signal input; pulse response frequency up to 250 Kpps.
- **Powerful, low repair rate:** uses imported industrial-grade components; mature and stable technology; fault-status indication and automatic protection for undervoltage, overvoltage, overcurrent, short circuit, open circuit, phase error, etc.
- **Compatible stepper motors:** 56, 60, 86 mm series.

Application Fields

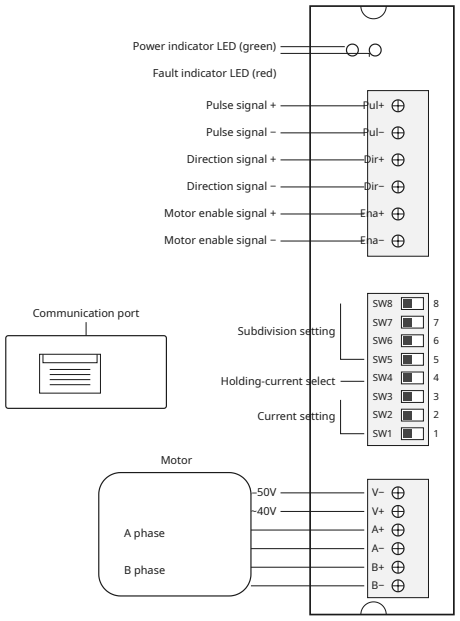
Suitable for automation equipment and instruments in all kinds of motion-control fields, for example: electronic machining and inspection, semiconductor packaging, laser cutting and welding, laser phototypesetting, packaging machinery, engraving machines, marking machines,

Indicator and Pin Function Description

Marking	Name	Notes
PWR	Power indicator LED	Lights green when powered on.
ALARM	Overcurrent / undervoltage indicator LED	Lights red when the current is too high or the voltage is too low.
PUL+	Pulse signal, positive terminal	Connect to the signal supply; +5 V

cutting machines, garment plotters, CNC machine tools and automated assembly equipment. It is the first choice for users in fields where low noise, superior high-speed performance and strong price/performance competitiveness are expected.

Driver Wiring Diagram



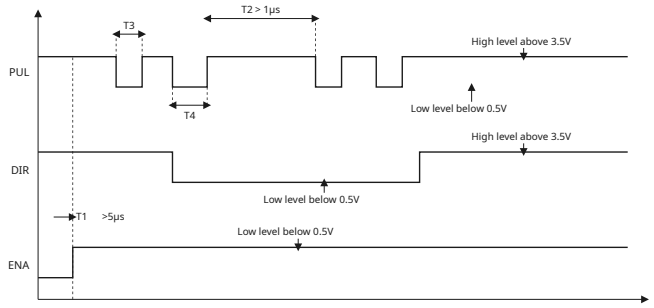
Signals (left), the SW1–SW8 DIP switch bank, the power terminals and the motor terminals (A+/A−, B+/B−), plus the communication port on the underside of the housing.

Running Current Setting

Peak current	SW1	SW2	SW3
1.0A	off	off	off
1.5A	on	off	off
2.0A	off	on	off
2.5A	on	on	off
3.0A	off	off	on
3.5A	on	off	on
4.0A	off	on	on
4.5A	on	on	on

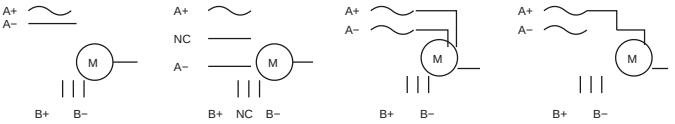
Input Signal Waveform Timing Diagram

Marking	Name	Notes
		to +24 V will all drive it.
PUL−	Pulse signal, negative terminal	Falling edge is active — the motor takes one step each time the pulse goes from high to low. Input resistance 220 Ω; requirement low level 0 to 0.5 V, high level 4 to 5 V pulse width 2.5 µs.
DIR+	Direction signal, positive terminal	Connect to the signal supply; +5 V to +24 V will all drive it.
DIR−	Direction signal, negative terminal	Used to change the motor's direction of rotation. Input resistance 220 Ω; requirement low level 0 to 0.5 V, high level 4 to 5 V pulse width 2.5 µs.
ENA+	Enable signal, positive terminal	Connect to the signal supply; +5 V to +24 V will all drive it.
ENA−	Enable signal, negative terminal	When active (low level) it cuts off the motor coil current; the driver stops working and the motor is in a free state.
V−	Supply, negative pole	DC24V ~ 50V AC20 ~ 40V
V+	Supply, positive pole	DC24V ~ 50V AC20 ~ 40V



Only T1 (> 5 µs) and T2 (> 1 µs) are given numerically in the original; T3 and T4 mark pulse-width intervals the datasheet leaves unlabelled. Level annotations: high level above 3.5 V, low level below 0.5 V.

Marking	Name	Notes
A− A+ B+ B−	Motor wiring	See the four winding options below.



4-lead | 6-lead (centre taps unused) | 8-lead — parallel (for high speed) | 8-lead — series (for low speed)

Microstep Subdivision Setting

Pulses/rev	SW5	SW6	SW7	SW8
default	on	on	on	on
400	off	on	on	on
800	on	off	on	on
1600	off	off	on	on
3200	on	on	off	on
6400	off	on	off	on
12800	on	off	off	on
25600	off	off	off	on
1000	on	on	on	off
2000	off	on	on	off
4000	on	off	on	off
5000	off	off	on	off
8000	on	on	off	off
10000	off	on	off	off
20000	on	off	off	off
25000	off	off	off	off

Note

1. The input voltage must not exceed DC50V / AC40V. When the driver is powered up, the green PWR indicator lights.
2. If the driver's fault indicator lights, check the motor wiring and the control signal lines. If it is a motor-wiring or other short-circuit fault, it must be powered off and on again to recover after the fault is cleared.
3. SW4 sets the holding-current function: **ON = Full Current**, **OFF = Half Current**. SW4 is set to OFF at the factory, which reduces heat generation in the motor and driver and improves reliability. The current is automatically reduced by 50 % 100 ms after the pulse train stops.
4. This driver has a single/double pulse selection function, settable through the communication port; the factory default is **pulse + direction**.

Source page furniture (not part of the datasheet):

Home > Stepper Drives > Two-Phase Stepper Drivers · (Click here to jump to Taobao to buy) · Product description

Previous: FYDM402T · Next: FYDM506T2 (discontinued) · Recommended related products

Friendly links: XFOYO – Alibaba International Store · Alibaba Chinese Store · Taobao Enterprise Store · Taobao Personal Store · English official website

Switchboard: 0755-27369350 · Phone: 13662209061 (Ms. Liu) · Email: sales@xfoyo.com

Address: Building D, Hedong Industrial Park, Hangcheng Industrial Zone, Xixiang Street, Bao'an District, Shenzhen

Copyright © Shenzhen XFOYO Electromechanical Co., Ltd. · ICP filing: 粤ICP备06058014号-4

Translation notes: page 2 of the source PDF is blank (a print artifact). The datasheet lists the compatible motors as the “56, 60, 86 mm” series; 56 mm is what is printed and the standard frame size is 57 mm (NEMA 23), so this is likely a typo in the original. Terminology: 细分 = microstepping, 锁机电流 = holding current, 脉冲/圈 = pulses per revolution.